

## POWER MARKETS

# Volume III Q&A Overview: Frontier Strategy

The third volume treats live market debates as decision problems: identify the durable mechanism, date the evidence, then decide what capability, risk, and capital commitment the claim implies.

# How do live market debates become decision-grade?

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|           |                                                                                                                  |
|-----------|------------------------------------------------------------------------------------------------------------------|
| QUESTION  | How can strategic work avoid becoming a bundle of current opinions?                                              |
| ANSWER    | Separate durable mechanisms from dated market conditions, then ask which decision boundary the evidence changes. |
| DEEPEN IN | Vol III Ch 1-14, especially Ch 12.                                                                               |

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## Durable mechanism

- Physics, network constraints, product design, risk allocation, asset limits, and governance requirements change slowly.
- These mechanisms decide what remains true when price levels or rules move.

## Dated condition

- Rule implementation, procurement volumes, market liquidity, price levels, competitor entry, and product depth change quickly.
- Dated evidence is useful when it is tied to a decision: build, buy, partner, defer, avoid, or gather evidence.

# What happens after BESS saturation?

|           |                                                                                                                                                                           |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| QUESTION  | How does a strategy change when many batteries crowd into the same products?                                                                                              |
| ANSWER    | Early scarcity rents compress when product depth, qualification rules, and competitor entry limit how much capacity can earn.                                             |
| ANCHOR    | Product depth is finite: reserve procurement volume, qualification rules, sustain requirements, activation patterns, and competitor entry determine how many MW can earn. |
| DEEPEN IN | Vol III Ch 1; Vol II Ch 8, Ch 9; Case C04.                                                                                                                                |

1

## Scarcity rent

- Early assets can earn from limited flexible capacity, shallow competition, and immature participation.

2

## Entry

- More batteries chase arbitrage, reserve capacity, balancing, and co-location value.

3

## Compression

- Product volume, sustain requirements, telemetry, activation patterns, and price response constrain the stack.

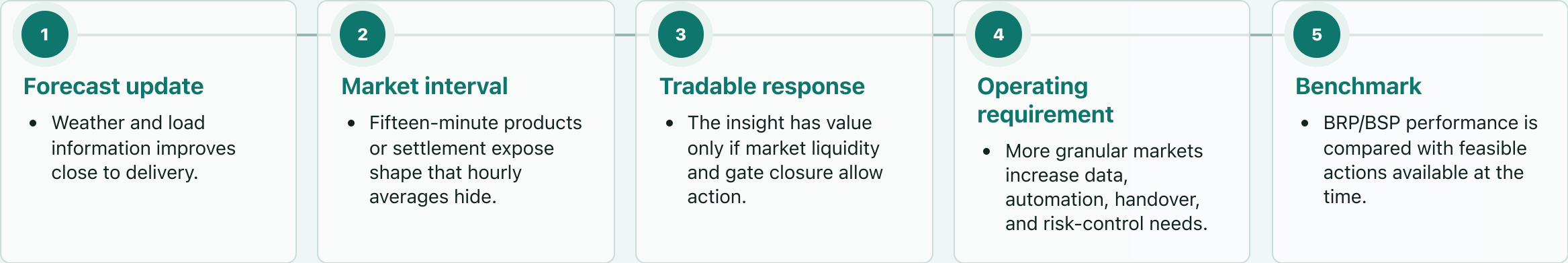
4

## Migration

- Value can move toward customer products, grid services, portfolio shaping, longer duration, or better governance.

# When does granularity create value?

|           |                                                                                                                                          |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------|
| QUESTION  | What changes when market intervals shorten and intraday liquidity improves?                                                              |
| ANSWER    | Granularity reveals sub-hourly shape and ramp value, but monetization requires liquidity, access, risk limits, and operational coverage. |
| ANCHOR    | 15-minute intervals create four decision/settlement blocks per hour instead of one.                                                      |
| DEEPEN IN | Vol III Ch 2; Vol I Ch 6, Ch 8; Vol II Ch 11, Ch 15.                                                                                     |



# When is congestion a strategy question?

|           |                                                                                                                         |
|-----------|-------------------------------------------------------------------------------------------------------------------------|
| QUESTION  | How can a grid bottleneck become a source of loss, option value, or product design?                                     |
| ANSWER    | Congestion changes site ranking, curtailment risk, connection timing, co-location value, and local flexibility options. |
| RELATION  | Curtailment loss scales as curtailed MWh × production-hour price, plus any contract or certificate consequences.        |
| DEEPEN IN | Vol III Ch 3; Vol I Ch 4; Vol II Ch 2; Case C03.                                                                        |

1

## Queue and connection

- Timing and reinforcement risk can dominate resource quality.

2

## Lost output

- Curtailment assumptions can move value materially at portfolio scale.

3

## Flexibility option

- Co-location, conditional access, local flexibility, or demand response can become valuable if rules allow it.

4

## Evidence pack

- Site screens need grid-right quality, tariff exposure, connection terms, curtailment history or forecast, and product eligibility.

# How visible is congestion in price?

QUESTION

ANSWER

DEEPEN IN

What changes when a market is zonal, nodal, or a hybrid of price signals and operational patches?

Market design decides where congestion appears: in the price, in redispatch, in tariffs, in curtailment, or in contractual basis exposure.

Vol III Ch 4; Vol I Ch 4, Ch 7; Vol II Ch 14.

## Design comparison

| Design         | Price signal                          | Operational patch                                | IPP implication                               |
|----------------|---------------------------------------|--------------------------------------------------|-----------------------------------------------|
| Zonal          | Average price inside a bidding area   | Redispatch, countertrading, tariffs, constraints | Hidden congestion and basis matter            |
| Nodal          | Local marginal value at network nodes | More granular dispatch signal                    | Location value is explicit                    |
| Hybrid reality | Rules plus exceptions                 | Network constraints still bind                   | Model price, grid, and rule exposure together |

# How hard is each clean-power promise?

|           |                                                                                                                                                     |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| QUESTION  | Why do higher-quality clean-power claims require more portfolio capability?                                                                         |
| ANSWER    | Each step up the promise ladder reduces the buyer's mismatch and increases the seller's need for data, firming, control, and residual-risk pricing. |
| RELATION  | Higher promise = more residual risk transferred from buyer to seller, plus stronger evidence and fallback obligations.                              |
| DEEPEN IN | Vol III Ch 5, Ch 7; Vol II Ch 4, Ch 5, Ch 6; Cases C01, C06.                                                                                        |

1

## As-produced

- Buyer receives the asset's physical shape and keeps much of the mismatch.

2

## Shaped or profile product

- Seller prices the transformation from production shape to customer need.

3

## Firmed clean supply

- Portfolio, storage, hedging, or market purchases support delivery in selected hours.

4

## Hourly evidence and 24/7-style claims

- Granular data, certificates, auditability, fallback rules, and stress-hour tail risk become central.

# What scarcity does longer duration solve?

|           |                                                                                                                                           |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------|
| QUESTION  | When is 4h, 8h, multi-day, or seasonal flexibility a different resource rather than a bigger battery?                                     |
| ANSWER    | Duration choice starts from the scarcity event: length, frequency, depth, location, and product value.                                    |
| RELATION  | A resource is adequate for an event only if usable MWh $\geq$ required MW $\times$ scarcity duration, after constraints and availability. |
| DEEPEN IN | Vol III Ch 6, Ch 13; Vol II Ch 9; Case C02.                                                                                               |

- Short-duration scarcity**
  - Intraday spreads, reserve sustain, solar evening shift, and short balancing needs can support one-to-four-hour assets.

- Longer-duration scarcity**
  - Longer peaks, multi-hour weather events, and profile products require extra MWh with a higher capital hurdle.

- Hard scarcity**
  - Multi-day or seasonal adequacy problems may need other technologies, demand response, firm contracts, or system-level procurement.

- Decision test**
  - More duration is justified when the marginal hours, constraints, and revenue quality support the extra capital.



# Why are data-centre products hard?

|           |                                                                                                                                                                                                      |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| QUESTION  | What makes a large load different from an ordinary renewable buyer?                                                                                                                                  |
| ANSWER    | A data centre creates a load-growth question, a grid-connection question, a reliability question, a flexibility question, and an evidence question at the same time.                                 |
| ANCHOR    | ENTSO-E's 2026 data-centre report cites roughly 15-30 GW of 2030 European short-duration/ramping flexibility need and a five-market estimate of 16.9 GW technical data-centre flexibility potential. |
| DEEPEN IN | Vol III Ch 7, Ch 5; Case C06.                                                                                                                                                                        |

## Product screen

| Screen  | Question                            | Risk if vague                    | Evidence                 |
|---------|-------------------------------------|----------------------------------|--------------------------|
| Load    | What is the hourly demand shape?    | Annual matching hides hard hours | Metered hourly profile   |
| Supply  | Which clean assets are deliverable? | Correlated or remote shortfall   | Asset and zone mapping   |
| Firming | Who covers residual hours?          | Uncapped market purchases        | Stress-hour tail model   |
| Control | Who can dispatch or curtail?        | Promise without rights           | Dispatch and data rights |
| Claim   | What is certified?                  | Reputational or compliance risk  | Granular evidence trail  |

# When can grid-forming capability become revenue?

|           |                                                                                                                                     |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------|
| QUESTION  | How does a technical inverter capability become a commercial asset?                                                                 |
| ANSWER    | Capability becomes value only through a local need, eligibility rule, measurement method, and settlement or avoided-cost mechanism. |
| DEEPEN IN | Vol III Ch 8; Vol I Ch 9, Ch 10.                                                                                                    |

1

## Capability

- Voltage support, fault behavior, frequency response, black-start support, and grid-forming behavior are distinct services.

2

## Need

- Local weak-grid or stability need matters more than generic capability claims.

3

## Route to value

- Value can appear as avoided delay, reduced curtailment, paid system service, compliance requirement, or future option.

4

## Proof

- Testing, telemetry, dispatchability, and market rules decide whether the capability can be counted.

# When is AI/ML commercially relevant?

|           |                                                                                                                                 |
|-----------|---------------------------------------------------------------------------------------------------------------------------------|
| QUESTION  | What has to be true before a model matters to energy management?                                                                |
| ANSWER    | Model uplift is commercially relevant when it changes a governed feasible action and can be benchmarked against an alternative. |
| RELATION  | Model value = decision uplift on feasible exposed volume – data, integration, governance, and error costs.                      |
| DEEPEN IN | Vol III Ch 9; Vol II Ch 10, Ch 11, Ch 15.                                                                                       |

1

## Data layer

- Rights, quality, latency, labeling, and history determine what can be learned.

2

## Decision layer

- Forecasting, bidding, dispatch, risk monitoring, and partner benchmarking have different objectives.

3

## Action layer

- Market access, liquidity, control rights, and risk limits decide whether the model can act.

4

## Governance layer

- Backtests, live benchmarks, model-risk charge, override rules, and PnL attribution separate tool value from theatre.

# Which trading capabilities remain internally accountable?

|           |                                                                                                                                                                 |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| QUESTION  | What can a renewable IPP outsource, and what remains internal accountability?                                                                                   |
| ANSWER    | Market access, analytics, execution, risk limits, data rights, and decision ownership are separate capabilities with different build, buy, and outsource logic. |
| DEEPEN IN | Vol III Ch 10, Ch 11; Vol II Ch 15, Ch 16, Ch 17; Case C07.                                                                                                     |

## Capability matrix

| Capability                   | Build                                 | Buy                       | Outsource                       |
|------------------------------|---------------------------------------|---------------------------|---------------------------------|
| Market access                | With scale and governance             | Platform or partner route | BRP/BSP execution               |
| Forecast and dispatch models | Own decision logic where strategic    | Use vendor components     | Audit the outsourced result     |
| Risk limits                  | Own internally                        | Benchmark tools can help  | Accountability remains internal |
| Data rights                  | Control architecture                  | Integrate systems         | Contract explicitly             |
| Execution                    | Build when volume and risk justify it | Hybrid possible           | Requires scorecards             |

# What does adequacy ask that annual MWh misses?

|           |                                                                                                                   |
|-----------|-------------------------------------------------------------------------------------------------------------------|
| QUESTION  | Why can high annual renewable output coexist with stress-period shortage?                                         |
| ANSWER    | Adequacy measures dependable contribution in scarcity periods; annual energy volume measures something else.      |
| RELATION  | Capacity credit = dependable MW contribution / nameplate MW, evaluated under system-specific scarcity conditions. |
| DEEPEN IN | Vol III Ch 13, Ch 6; Vol I Ch 3, Ch 13.                                                                           |

## Capacity credit

- Dependable contribution depends on system conditions, correlation, outage risk, and scarcity duration.
- A resource's contribution changes as the system mix changes.

## Scarcity duration

- A two-hour battery may be valuable in one scarcity regime and insufficient in another.
- Adequacy questions push analysis toward diversification, firming products, demand flexibility, market purchases, and capacity mechanisms.

# How are scarcity and market power separated?

QUESTION

ANSWER

DEEPEN IN

When are high prices useful signals, and when do they require market-monitoring evidence?

Scarcity, conduct, and market design have to be separated before a portfolio treats high prices as investable upside.

Vol III Ch 14, Ch 13; Vol I Ch 7.

## Evidence map

| Object       | What to examine                                                 | Portfolio implication                    |
|--------------|-----------------------------------------------------------------|------------------------------------------|
| Scarcity     | Supply margin, demand response, reserve level, activation       | Flexibility and investment signal        |
| Market power | Withholding, local dominance, constrained alternatives          | Conduct and reputational risk            |
| Rule design  | Caps, mitigation, monitoring, redispatch, settlement            | Regulatory-change exposure               |
| Evidence     | Bids, constraints, liquidity, rule compliance, monitor findings | Treat upside and conduct risk separately |

# What makes a frontier claim decision-grade?

|           |                                                                                                                                                     |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| QUESTION  | How does a strategy memo handle uncertainty while staying decision-grade?                                                                           |
| ANSWER    | A frontier claim becomes useful when it names the mechanism, dated evidence, capability need, exposure, and trigger that would change the decision. |
| DEEPEN IN | Vol III Ch 12, Ch 1-14; Case C07.                                                                                                                   |

1

## Mechanism

- What physical, market, or contractual force creates value?

2

## Evidence

- What is dated, sourced, and still uncertain?

3

## Capability

- What data, rights, models, partners, and controls are required?

4

## Exposure

- What can go wrong financially, operationally, legally, or reputationally?

5

## Trigger

- What new evidence would change the decision?